The Interior Stability Barrier

Why Stage-3 Requires a New Structural Input to NFC
A Proposed Paper BU for the Dream-Paper Suite

Evan Thomas Kotler

March 20, 2026

Abstract

The NFC monograph (Open Problem 36.27, Conjecture 36.28, Remark 36.8) makes an explicit negative statement:

> **"Stage-3 is not within reach of the current NFC framework. The quadratic bound $D_{\text{floor}}(U_n) \leq C_* a_n^2$ requires a new structural input to NFC that is not available: a mechanism by which the certified-event density decreases under lattice refinement. This is the precise and honest statement of the Stage-3 frontier."**

The reason is specific. The interior certified-event rate is:

$$\eta_n := \frac{N_{\text{cert}}(\text{int}(U_n), t)}{T_{\text{cov}}(U_n) \cdot |\text{int}(U_n)|} \approx \frac{2}{3} \quad \text{for all } n.$$

This rate is *scale-independent*: WL-1 stabilization fixes the collar alphabet at depth ρ^*; NFC lattice refinement is a relabeling at fixed resolution, not a dynamical renormalization; the $K_0 = 7$ collar walk operates on the same 9-class graph at every scale; LS-2 locality makes the per-step certified-event rate local and therefore scale-independent.

Paper BT's H-CODIM2 (codimension-two carrier concentration) is equivalent to the statement that $\eta_n = O(K_0^{-n/2})$, i.e., the interior rate decays. The monograph explicitly states: this does NOT hold under the current axiom set, and Conjecture 36.28 names it the Renormalization Flow Conjecture — currently **not supported**.

Paper BU formally registers this barrier, identifies the precise new structural input required, and provides the first honest formulation of what a dynamical renormalization group (dRG) on the NFC collar walk would need to look like.

Contents

1 The scaling obstacle — formal statement **2**

2 What new structural input is required **2**

3 The dRG as a new bridge paper **3**

4 Honest status of H-CODIM2 **3**

5 Status ledger **4**

6 Conclusion **4**

1 The scaling obstacle — formal statement

Open Problem 1.1 (NFC Open Problem 36.27 — Interior Stability)**.** Define the interior certified-event rate:

$$\eta_n := \frac{N_{\text{cert}}(\text{int}(U_n), t)}{T_{\text{cov}}(U_n) \cdot |\text{int}(U_n)|}.$$

Stage-3 Sub-Obligation (Interior Stability): Prove or disprove: $\eta_n = O(K_0^{-n/2})$.

If this holds, then $N_{\text{cert}}(\text{int}(U_n), t) = O(|\partial U_n|)$, and together with the boundary-certified-event bound (Lemma 36.19 of the monograph), Strong BA follows.

Current status: Does NOT hold under existing axioms.

Theorem 1.2 (Scale-independence of interior rate — from monograph)**.** *Under the current NFC axiom set (WL-1, PASS, Phase-A, LS-2, IC-1):*

$$\eta_n \approx \frac{2}{3} \quad \textit{for all } n.$$

No scale-dependent Phase-A contraction exists. Stage-3 is not within reach.

Proof. By WL-1 stabilization (Paper BO Theorem 2.1), the collar alphabet stabilizes at depth ρ^* and produces $K_0 = 7$ fixed color classes. NFC lattice refinement $U_n \to U_{n+1}$ is a relabeling at this fixed resolution, not a dynamical renormalization of the collar walk.

The $K_0 = 7$ collar walk operates on the same 9-class graph at every scale n. For the background class C_8 (weight $(0, 0)$, size 25 in the NF2 structure), exactly 2 of 3 outgoing transitions leave C_8, giving $\eta_n \approx 2/3$ scale-independently.

By LS-2 locality (Paper A Definition 5.3), the per-step certified-event rate at any interior site is determined by the local $K_0 = 7$ class transition structure, which is scale-independent. The $\tau \leq 4$ Phase-A ceiling holds at all scales and does not tighten under refinement. □

Conjecture 1.3 (NFC Conjecture 36.28 — Renormalization Flow)**.** *For Stage-3 to close, the interior certified-event rate must decay under refinement:*

$$\eta_n = O(K_0^{-n/2}).$$

Current structural status*: not supported. No scale-dependent Phase-A contraction exists within the current axiom set.*

2 What new structural input is required

Definition 2.1 (Dynamical renormalization group (dRG) on the collar walk)**.** A *dynamical renormalization group* (dRG) on the NFC collar walk is a mechanism $\mathcal{RG}$: collar walk at scale $n \mapsto$ effective collar walk at scale $n + 1$ such that:

(i) $\mathcal{RG}$ is not merely a relabeling: it genuinely changes the effective transition probabilities between collar classes;

(ii) $\mathcal{RG}$ suppresses the certified-event rate: the effective transition rate at scale $n + 1$ is strictly smaller than at scale n;

(iii) $\mathcal{RG}$ is compatible with the NFC architecture: it respects LS-2 locality, Phase-A stability, and the IC-1 capacity bound.

Remark 2.2 (Why a dRG is the right new input). The monograph is explicit: what is needed is "a DYNAMICAL renormalization group acting on the collar walk itself (not just on the labeling), so that at scale $n+1$ the effective dynamics suppresses certified events that were permitted at scale n."

A dRG would change the effective collar walk at finer scales, not merely relabel it. This is the NFC analogue of Wilsonian renormalization: at coarser scales, the effective degrees of freedom are fewer and the certified-event rate is lower, because fine-scale fluctuations have been integrated out.

The push document (iterations 165–170) shows that such a mechanism would give the codimension-two concentration: the effective fine-scale dynamics would confine residual defect-floor carriers to lower-dimensional structures, exactly as vortex filaments in 3D fluid dynamics concentrate on sets of dimension 2.

3 The dRG as a new bridge paper

Open Problem 3.1 (NFC Bridge Paper H-CODIM2 / Paper BV: dRG Construction)**.** Construct a dynamical renormalization group $\mathcal{RG}$ on the NFC collar walk satisfying Definition 2.1, and prove that it induces the interior rate decay $\eta_n = O(K_0^{-n/2})$.

Candidate mechanism: The Lieb–Robinson bound (Paper BD) already shows that under H-LR, information propagation in $\mathfrak{A}_\infty$ is causal and finite-speed. At finer scales, the effective number of independent degrees of freedom per unit volume decreases because the Lieb–Robinson correlation length $\xi_{\mathrm{LR}} = \mu^{-1}$ is fixed while the lattice spacing $a_n \to 0$.

Specifically: at scale n, each interior site interacts with $O((\xi_{\mathrm{LR}}/a_n)^3)$ neighbors. As $a_n \to 0$, this count grows, but the effective collar-break rate per site decreases because the Lieb–Robinson bound exponentially suppresses the influence from distant sites. The effective interior rate at scale n would be:

$$\eta_n^{\mathrm{eff}} \approx \eta \cdot e^{-\mu\, a_n/\xi_{\mathrm{LR}}} \;=\; \eta \cdot e^{-\mu a_n} \;\ll\; \frac{2}{3}$$

for small a_n. This would give $\eta_n^{\mathrm{eff}} = O(a_n) = O(K_0^{-n/2})$, exactly the decay needed for Conjecture 1.3.

What remains: Formalizing this heuristic into a theorem requires identifying the fluid-observable sub-algebra of $\mathfrak{A}_\infty$ (H-LR) and showing that the effective collar-break rate at scale n is determined by the fluid sector's Lieb–Robinson propagation speed. This is H-CODIM2 + H-LR in concert.

4 Honest status of H-CODIM2

Remark 4.1 (Correction to Paper BT). Paper BT assumed H-CODIM2 as a hypothesis and built the quadratic floor theorem upon it. The present paper clarifies that H-CODIM2 is currently:

- **Motivated**: by the Lieb–Robinson decay mechanism (heuristic above);
- **Consistent**: with the NFC axiom set (no contradiction is known);
- **Not provable**: from the current NFC axioms (Theorem 1.2 shows the naive rate is scale-independent);
- **Equivalent** to Conjecture 36.28 (Renormalization Flow Conjecture) + H-LR; and
- **Requiring** a new structural input: either a dRG mechanism or an identification of the fluid-observable sector with a sub-algebra of $\mathfrak{A}_\infty$ in which the Lieb–Robinson bound provides effective rate suppression.

The NS Prize question, in the most precise NFC language, is:

> *Does the Lieb–Robinson bound, applied to the fluid-observable sub-algebra of $\mathfrak{A}_\infty$, induce an effective dynamical renormalization group that suppresses the interior certified-event rate from 2/3 to $O(K_0^{-n/2})$?*

5 Status ledger

Label	Claim	Status
BU1	Interior rate $\eta_n \approx 2/3$ (scale-independent)	UNC (monograph)
BU2	Stage-3 not in reach of current NFC axiom set	UNC (monograph)
BU3	Conjecture 36.28 (Renormalization Flow)	OPEN
BU4	dRG candidate mechanism (LR-induced rate suppression)	OPEN
BU5	H-CODIM2 $\equiv$ Conjecture 36.28 + H-LR	OPEN
H-CODIM2	Residual carriers on codim-2 support	OPEN (not in current axioms)
H-LR	Lieb–Robinson for fluid sector	OPEN
NS Prize	Global smooth NS solutions	**COND (H-CODIM2 + H-LR)**

6 Conclusion

The NS Prize question is, in the suite's most precise language, the question of whether a dynamical renormalization mechanism exists for the NFC collar walk — one that suppresses the interior certified-event rate as the lattice spacing $a_n \to 0$.

The full L_∞ Lieb–Robinson bound (Paper BD) is UNC. H-LR asks whether this bound applies to the fluid-observable sector. H-CODIM2 asks whether that localization is strong enough to induce effective rate decay $\eta_n \to 0$. Together (H-LR + H-CODIM2), the NS Prize is conditionally resolved by Paper BS/BT.

What is currently unknown is whether the NFC structure, specifically the Lieb–Robinson propagation in $\mathfrak{A}_\infty$, provides a sufficiently strong suppression mechanism for interior certified events at fine scales. This is a precise and bounded open question — not a vague frontier, but a quantitative question about the relationship between the Lieb–Robinson length μ^{-1}, the lattice spacing a_n, and the effective collar-break rate at interior sites.